Excursion: E-mission-head — make the mission HEAD a typed object

♡ vitals (computed from this document)

confidence 0.33 · xenotype 89% · exotype 00011100

⊗IF ⊗HOWEVER ✓THEN ✓BECAUSE

alive but wobbly: xenotype completeness 0.89; strong anchors ['THEN', 'BECAUSE'], weak anchors ['IF', 'HOWEVER'].

generated 2026-06-10T18:16:29 by `head_exotype_probe.py`

Date: 2026-06-10 **Status:** :greenfield — HEAD only, by design. This excursion is itself the demonstration: it is being developed live with the `*mission-overview*` panel open, so the Scratch-Map builds up from scratch as each phase section lands. **Spawned:** from E-scope-audit session 1 (W10 + W11), Joe + Fable, unclocked.

HEAD (Joe, 2026-06-10, verbatim sense)

A mission's HEAD is doing two things at once.
First, in a Jazz or Free Jazz sense, the HEAD is the **theme around which the whole mission is improvised** — ideally in a fully automated way, by following the **patterns of improvisation** that Joe intends to learn from the futon5 work. We are not there yet.
Second, each mission is a **virtual peripheral**, and the HEAD is an **AIF head**: it should be** mappable to the usual AIF terminal vocabulary** — priors, preferences, observations, policies — which would allow us to understand, in typed terms, **what it means to satisfy the requirements of the mission**.
Concretely, the HEAD should therefore carry an extraction of **concepts** — per the **Interest Network** — named entities like "storage", "agents", "patterns", "meme graph", "futon3b", "pattern cascades", obtainable just by *reading the HEAD*. The current kernel extraction is unigram-grade; the upgrade is Interest-Network-grade extraction over the mission corpus, feeding the mission-mode concept lane that already exists. The satisfaction-conditions half connects to E-ground-G: a HEAD mapped to AIF terminal vocabulary is what grounded-G ultimately discharges against.

1. IDENTIFY

The gap (Joe, 2026-06-10, verbatim sense)

Missions right now are treated as **project management tools** — fine, but that ignores that project management tools *are* "peripherals" in the peripheral-spec sense: **constrained execution envelopes**. Gantt charts, Google Calendars, project plans — it checks out. The problem: those are all very boring, and **Missions are way cooler — because missions are fun.** They give you a sense of **purpose, not just progress**. And we don't know how they will develop until we run them — that is the improvisatory part. `mission-lifecycle.md` gives the rough shape, but (for example) we don't always know in advance how we will VERIFY a design — we can do it in a clever way depending on how the mission shows up. Unlike a Gantt chart, **we can't know in advance.**
The gap as experienced inside futonic development: a mission's HEAD is both the **thematic statement of intent** *and*, implicitly, **a small essay describing the mission's purpose**. With those ingredients we should be able to build an **AIF model of the mission** (per `futon5/README-aif+.md`), so the mission becomes an **AIF "lifeform" the moment it is seeded** — we could see whether it is strong or weak, missing context, what its boundaries are, all at the start; and by the time it is built it should look *healthy*, the structural weaknesses ameliorated. Benefits: **#1 automation, #2 quality, #3 integration** with the AIF-hierarchies modelling.

Where the gap is already palpable in the stack (evidence pass, 2026-06-10)

1. **`futon2/holes/M-aif-head.md` — COMPLETE 2026-03-15: "AIF Heads for Every Peripheral (Mission Peripheral First)."** The strongest evidence: this gap was *a closed mission three months ago* ("organs without a body — no shared generative model"), yet the gap is still felt. Either it built something narrower than the felt need, or it built the right thing and the practice never inhabited it (the meme.arrow failure mode: revive-vs-replace applies). **First question of MAP: what did M-aif-head actually deliver, and why doesn't today's mission tooling feel it?**
2. **Missions already ARE peripherals in code** — `futon3c/src/futon3c/peripheral/mission.clj` + `mission_backend.clj` (the Mission Peripheral wraps mission-domain operations). The envelope exists; what's missing is the *generative model inside it* — the peripheral constrains execution but holds no model of the mission's own health.
3. **Hand-built AIF models of missions already exist** — `futon5/data/missions/*-exotype.edn` (aif2, coordination, social, evidence-landscape). These are exactly "AIF model of a mission" — but hand-authored, for four missions, not *seeded from the HEAD*. The gap = the missing compiler from HEAD → exotype.
4. **`futon3/library/futon-theory/mission-interface-signature.flexiarg`** — the single most-attested pattern in the corpus (≈236 attestations in `pattern-attestation.json`): mission-as-typed-interface is already the stack's strongest habit, waiting to be formalized.
5. **The capability star-map** (`futon0/.../M-capability-star-map.graph.edn`) types missions by `:scope`/`:produces` — a proto-AIF signature (what it senses / what it acts on) — and the WM already does EFE over it. A seeded-at-birth lifeform model is what that machinery is starving for (cf. WM "starved of input sources").
6. **The watcher already parses the essay half** — `parse-mission-md` extracts `:mission/summary` (first paragraph) into substrate-2 `mission-doc` hyperedges, where it sits **unused for any model**. The HEAD's raw material is in the store today.
7. **E-ground-G + this excursion's HEAD**: grounded-G needs satisfaction conditions to discharge against; a HEAD mapped to AIF terminal vocabulary *is* those conditions. The two excursions meet exactly here.
8. **mission-mode/scope-audit session 1** (`futon6/holes/E-scope-audit.md`): the lifecycle is now *visible* (phases, ghosts, in-passing closures) but purely descriptive — the panel can show a mission's shape, and has no notion of its *strength*. The UI is a lifeform-viewer with no lifeform.

IDENTIFY exit

The gap is real, felt, and **eight-times palpable**; the decisive unknown is item 1 (what M-aif-head delivered vs. what is lived-in). → MAP should open with the M-aif-head autopsy, then inventory the exotype schema against what a HEAD actually contains.

2. MAP (2026-06-10) — autopsy + the futon5 improvisation apparatus

2.1 M-aif-head autopsy: built, 0% wired — the head never attended the birth

M-aif-head (COMPLETE 2026-03-15, futon3c commit `51ae8eb`) delivered **real, well-designed code**:

- `futon3c/src/futon3c/aif/mission_head.clj` (242 ln) — `MissionAifHead`, EFE + softmax action selection over a bounded action space;
- `futon3c/src/futon3c/aif/observe.clj` (186 ln) — **ten observation channels** (phase-progress, obligation-satisfaction, structural-law compliance, prediction-divergence, ...) — i.e. exactly the "is this mission strong or weak" vocabulary this excursion wants;
- `futon3c/src/futon3c/aif/invariant.clj` (92 ln) — C9: *every peripheral has an AIF head*, as an executable check;
- the `:on-cycle-complete` hook installed in `peripheral/cycle.clj:159`.

And none of it is alive: zero callers for any export; the hook is installed but no peripheral config ever passes a callback; `mission-domain-config` (`peripheral/mission.clj:144`) was never updated; the `validate-phase-advance` call site uses the 2-arity (no head, no state). Three handoffs never landed (H-1 AifHead protocol in futon2 — claimed, file absent; H-2 inventory loader; H-8 portfolio effect sink). **The mission designed the head but never wired it into the Mission Peripheral's birth ceremony.** The felt gap = the distance between "the peripheral IS an AIF organism" (promised) and "AIF scaffolding sits on disk unexercised" (actual). Same failure mode as meme.arrow (built
neq inhabited). **Verdict: REVIVE** — the architecture is sound and load-bearing (cited by later missions); the work is integration, not invention.

2.2 futon5: the patterns-of-improvisation apparatus (ACTIVE — latest commit 2026-06-09)

- **The 256 "universal" patterns exist as the II Ching**: 64 hexagrams × 4 energy levels = 256 physics rules, library at `futon3/library/iiching/` (skeleton: `futon3/library/iching/`, 64 files) — *the pattern library lives in futon3, the play lives in futon5*, as stated. Machinery: `futon5/256ca.el` (2014 meta-CA, 256 sigils), `src/futon5/hexagram/lift.clj` (6×6 exotype matrix eigendecomposition → 6 eigenvalue signs → hexagram lines).
- **The bridge is built**: `scripts/pattern_exotype_bridge.py` (Feb 2026) maps **all 791 library patterns → 8-bit exotypes + 36-bit xenotypes** via MiniLM(384) → PCA(32) + Ridge, trained on 320 anchors; held-out hexagram bit-accuracy 57.9%, Hamming 3.37±1.33; output `resources/pattern-exotype-bridge.edn`.
- **The ants↔CA↔CT triangle is working code**, with real transformations: pattern→exotype (bridge), exotype→MMCA rule (`mmca/exotype.clj`), exotype→ant-AIF deltas (`cyber_mmca/core.clj`), pattern→CT wiring diagram (`scripts/pattern_to_wiring.clj`; 6 compiled wirings in `holes/e-warranted-play/wiring/`, 2026-06-09), xenotype evaluation registry (`xenotype/interpret.clj`, 17 scoring modes incl. filament/EOC).
- Known tension from the play: coupling × diversity are anti-correlated (Pareto at gen 12) — relevant later if mission-lifeforms get scored.

2.3 The conjecture: mission scopes as early evidence of the 256

Assessment: **credible, unproven, and now testable.** The shape alignment is real — a 36-bit xenotype is a *situational binding* (where/what/why/who a pattern applies), which is what a mission scope *is*; pattern IF/HOWEVER/THEN/BECAUSE parallels the eightfold phases. What does not yet exist: any explicit hexagram↔scope-type mapping artifact, and the scope detector does not ingest futon5's enumeration. The honest framing: the binder vocabulary (11 types) is the *empirical* end and the II Ching (256) the *universal* end; the test is whether observed scope-instances cluster onto a small stable set of binding-shapes that embed near hexagram anchors in the bridge's space.

2.4 The synthesis MAP hands to DERIVE (the hot finding)

The HEAD→AIF-model compiler partially exists already. A mission HEAD is text; `pattern_exotype_bridge.py` embeds arbitrary pattern text into an 8-bit exotype + 36-bit xenotype *today*. So the v0 of "mission becomes an AIF lifeform the moment it is seeded" is a composition of parts on disk:

> *HEAD text → (bridge) → exotype/xenotype → hexagram lift → run/score (interpret.clj, 17 modes incl. EOC) → **health-at-birth readout** → seed `MissionAifHead`'s 10 observation channels (the revived M-aif-head) → display in mission-mode (the lifeform-viewer that's waiting).*

Every arrow except the first and last two exists as working code. DERIVE's question is not "can this be built" but **which composition is the honest v0** — and what "health" means at seeding time (strong/weak, missing context, boundaries) in terms of the ten observe.clj channels + the xenotype scoring modes.

2.5 This mission's own wiring diagram — written into the meme store (2026-06-10)

Joe's steer: we should build a **wiring diagram per mission** — the cascade→sorry→wiring-diagram evidence discipline being developed with claude-3 — and hold mission development to **the same standard as the AIF learning loop** ("did it help?", realized closure, no laundering). *"We" (Joe) are the Cyborg version of that same learning loop.* No time like the present: §2.4's chain **is** a wiring-diagram-with-gaps, so it now lives as arrows in the live futon3a meme store (`futon3a/meme.db`), scope-tagged **`diagram/E-mission-head`** (the R3 named-composite container isn't built yet; the tag is the v0 grouping):

- **4 × `:constructed`** (mode `:construction`, evidence in rationale): head-text→exotype/xenotype (the bridge); encoding→hexagram (lift); encoding→evaluation-score (interpret.clj); aif-head-code→ten-channels (M-aif-head's dormant deliverable).
- **3 × `:open` — the mission's sorries, RHS specified: score**→health-at-birth-readout**; readout**→seeded-beliefs **(the unwired birth ceremony, §2.1); readout**→mission-mode-lifeform-lane** (the viewer awaiting its lifeform).

Closing this excursion honestly = `promote!`-ing those three arrows `:open→:constructed` with real constructions — measurable in the store, per the E-ground-G standard (closure events, not assertions). Note in passing: minting required applying the meme schema's own pending `advances_cap` migration (schema.clj's exact DDL; locked WM-policies contract §4a) — the store's first non-claude-4 write found the seam honest.

3. DERIVE — opened with a candidate runner (2026-06-10)

3.1 The probe: this HEAD, through the default chain

`futon5/scripts/head_exotype_probe.py` (new; the **candidate runner**) runs the bridge's *default* representation on a mission HEAD — deliberately, so representation failures surface honestly (Joe's prediction: "our default way of doing that wouldn't actually work well"). Run on this excursion's own HEAD (1190 chars):

- **exotype `00011100` → **rotation=0, match-threshold=1, invert-on-phenotype=true, mix=0 — but** bit-confidence 0.29 (LOW)**: per-bit proba `0.13 0.44 0.43 0.61 0.56 0.55 0.25 0.32` — four of eight bits are near coin-flips. Three bits are confident (b0=0, b6/b7≈0); the middle of the rule is noise.
- **nearest anchors are all weak** (cos ≈ 0.35; nothing close): exotype-242/243, **hexagram-38 Kui (Opposition/estrangement)** and **hexagram-49 Ge (Revolution/molting)** — poetically apt for a mission about a felt gap and missions-becoming-lifeforms, but at Δcos 0.002 the poetry is not signal.
- **xenotype (36-bit): not computable at all** — the bridge derives it from IF/HOWEVER/THEN/BECAUSE sections, and a HEAD has none. Plus a length mismatch (1190-char prose vs short pattern texts the anchors were trained on).

Prediction confirmed: the default representation is weak here — and now that's a measured fact with a runner, not a vibe.

3.2 The representation finding (the convergence)

The failure is *structural*, and the fix is something this excursion already wanted for independent reasons: **the xenotype path requires sectioned text, and W11's HEAD→AIF-terminal-vocabulary mapping IS a sectioning of the HEAD**:

> *intent/theme ≈ IF (priors) · the felt gap ≈ HOWEVER (observation divergence) · the move ≈ THEN (policies) · the warrant ≈ BECAUSE (preferences/-values)*

So the representation change that makes the bridge work on HEADs and the AIF mapping that makes satisfaction conditions typed are **the same move**: parse (or author) the HEAD into AIF-terminal sections, embed section-wise, and the 36-bit xenotype — the *situational binding*, i.e. the lifeform's boundary — becomes computable. v0.1 of the runner = section the HEAD, re-run, compare bit-confidence against today's 0.29 baseline (falsifiable: if sectioning doesn't raise it, the weakness is deeper than structure).

3.3 The recast (Joe steer, 2026-06-10): HEAD as design pattern — Golemization

> *"The HEAD should be recast as a design pattern. That's exactly what we did for AIF ants — we installed sigils (II Ching patterns) into them, to **Golemize** them."*

The ants precedent (`futon5/src/futon5/cyber_ants.clj`, `cyber_mmca/core.clj`): an ant is animated not by prose but by an installed **sigil** — the pattern already compiled to its 8-bit animating word, translated into AIF deltas. The Golem's shem. Same move for missions: author (or project) the HEAD in the **IF/HOWEVER/THEN/BECAUSE** pattern form, and the bridge's section path gives the full 36-bit xenotype — the sigil to install in the mission peripheral at birth. **This excursion's HEAD, recast:**

IF: you want missions to be living things — purpose, not progress — improvised around a theme the way a jazz head seeds a performance, ideally automatable by following learned patterns of improvisation.

HOWEVER: missions are currently treated as project-management tools — Gantt-grade constrained execution envelopes; the HEAD is prose; nothing can read a mission's strength, weakness, missing context, or boundaries at seeding time; satisfaction of the mission's requirements is untyped.

THEN: recast the HEAD as a design pattern and compile it — install its sigil (exotype + xenotype via the pattern-exotype bridge) into the mission peripheral at birth, Golemizing the mission: an AIF model seeded from the theme, with health observable from day one.

BECAUSE: each mission is a virtual peripheral and the HEAD is its AIF head; a HEAD mapped to AIF terminal vocabulary yields typed satisfaction conditions (what grounded-G discharges against); and the ants showed that installing II-Ching sigils into AIF agents is sufficient to animate them.

3.3.1 v0.1 RESULT — the Golemization run (2026-06-10)

The probe's pattern-mode ran on §3.3's recast (same projector, section-wise):

section	bits	conf	nearest anchor	cos
IF	`00011100`	0.28	hexagram-49 **Ge** (molting/revolution)	0.284
HOWEVER	`00011001`	0.36	hexagram-13 **Tongren** (fellowship)	0.306
THEN	`10101100`	0.32	iiching/exotype-248	**0.454**
BECAUSE	`11001000`	0.37	hexagram-31 **Xian** (mutual influence)	**0.461**

xenotype-32 `00011100·00011001·10101100·11001000`, mean-conf 0.33 (baseline 0.29). Findings, honestly weighted:

1. **Qualitative unlock confirmed: the sigil now exists.** The xenotype was *uncomputable* before the recast; the Golem has a complete shem.
2. **Bit-confidence lift is modest** (+0.04). Sectioning helps; it does not transform the projection. (Caveat: v0.1 reused the whole-pattern projector per section; the bridge's official path trains per-section projectors — some headroom there.)
3. **The real signal moved into anchor proximity, asymmetrically**: THEN and BECAUSE land at cos ≈ 0.46 — a third stronger than the whole-text best (0.349) — while IF/HOWEVER stay weak (≈ 0.29). Reading: **the actionable half of a HEAD (move + warrant) projects into pattern-space well; the intent/gap half is mission-specific prose the pattern-grain anchors don't cover.** Supports next-car 3: HEAD-grain anchors (missions with known outcomes) for the IF/HOWEVER half.
4. The oracle stayed coherent: intent→*molting*, gap→*fellowship*, warrant→*mutual influence*. Recorded as flavor, not evidence.

3.3.2 The lifeform package: `E-mission-head.aif.edn` (2026-06-10)

Joe: an AIF+ lifeform should ship with a **`.aif.edn` package** — they are supposed to be able to **contend in a Calculemus battle** with other AIFs. So this mission now has one (sibling file, canonical AIF+ v2 shape per `futon5a/holes/stories/leaf-argument.aif.edn`): the **sigil** (exotype + xenotype-32 + per-section anchors), **birth vitals** (`:health` — completeness 32/36, bit-confidence 0.33, the proximity asymmetry, construction-state 4/3), the recast as spine+claims, four **falsifiability nodes** (F1 discharged-PASS = the v0.1 result; F2 decorative-readout, F3 uninformative-sigil, F4 intent-not-typeable — all spec-only), and support/falsifies edges.
Contendability proven: staged via `futon5a/scripts/run_aif2_contest.clj` ("Calculemus!") against `leaf-argument.aif.edn`. Verdict `indeterminate` — correctly: no cross-edges were authored, and the package's structural metrics match the canonical exemplar's. A real bout = authoring attack edges against an opponent's claims; that is a deliberate future move, not tonight's. (Runner quirk found: `(drop 1 *command-line-args*)` eats the first flag under `bb`; invoke with a dummy leading arg.) Also recorded: **Skolemization as an alternative Shem-computation** — the three `:open` arrows are existentials; naming their witness functions is another way to write the animating word. *"Skolemization: tired → Golemization: wired."*

3.4 DERIVE next cars (superseded in part by §4 amendments)

1. **v0.1 runner**: section this HEAD per §3.2 (hand-sectioned first — cheap), re-run, report Δconfidence + whether a xenotype materializes.
2. If Δ is real: the `health-at-birth-readout` open arrow gets its first component (bit-confidence + anchor-proximity + xenotype-completeness as health channels), and the HEAD-authoring convention gains a typed template.
3. Anchor-set question for later: 1190-char HEADs may want HEAD-grain anchors (missions with known outcomes) rather than pattern-grain ones — the mission-corpus equivalent of the 320-anchor set.

4. ARGUE (2026-06-10) — the lifeform's self-argument: why should I exist?

Form (a first): this ARGUE was conducted *as a Calculemus bout* — the excursion's own `.aif.edn` lifeform contending against a **steel-manned adversary package** (`E-mission-head-contra.aif.edn`) whose every claim is true: the **graveyard base-rate** (M-aif-head 0% wired; meme.arrow 5 months empty), the **Goodhart surface** (health-at-birth invites sigil-gaming), the **jazz objection** (typing the HEAD Gantt-ifies the very thing that makes missions fun), and the **thin-signal objection** (+0.04 mean-confidence). Runner: `futon5a/scripts/run_aif2_contest.clj`.
Round 1 — the arena caught a real flaw: 'a-thesis-attacks: 0, b-thesis-attacks: 1'. The lifeform rebutted every contra pillar but never struck the contra's thesis — it argued like a defender, not a contender (undercutting supports
neq defeating the conclusion).
Round 2 — the thesis-strike, and symmetry: added `n0 →attacks→ nC0`: *remaining prose is not the safe default — it is the status quo that already failed, measurably* (the eight-times-palpable gap; the WM starving; untyped satisfaction giving grounded-G nothing

to discharge against; M-aif-head dying **invisibly** precisely because nothing measured its non-wiring). Result: **Symmetry: mutual** (thesis 1:1, 6 vs 5 cross-edges).
What the contra earned — amendments codified in the package:

- **A1 (from nC2/Goodhart):** the health readout **never gates birth, never enters any optimization target** (G, EFE, fitness) — observation + sparse audit only. The Xbox-achievement discipline, now an invariant; violating it triggers the lifeform's own F2.
- **A2 (from nC1/graveyard): wire-or-die** — if the three `:open` arrows in `diagram/E-mission-head` are not promoted with real constructions within the next two working sessions on this thread, the lifeform self-declares `:moribund` in its own package. The graveyard claim is load-bearing, not dismissed.

Verdict: `indeterminate` — and that is the honest ARGUE exit. Both sides' falsifiability nodes are `:spec-only`; existence is warranted *conditionally*: the lifeform lives exactly as long as F2–F4 stay undischarged-against-it and A2's clock is respected. The strongest defense made (nBEC vs nC3) stands as the excursion's core position: *a jazz head IS a typed object — changes, key, form — and the solos are wild because the theme is stable; the ants were animate BECAUSE of their sigils.*
ARGUE exit → operator ratification. Per house discipline the phase closes on Joe's call: ratify the amendments (esp. A2's two-session clock) and the conditional verdict, or send the lifeform back to the arena.

4.1 Plain-language argument (the version anyone can read)

When we start a piece of work, we write a short statement of what it is for. Right now that statement is just words: nothing else in the system can read it, check it, or use it. So nobody can tell, at the start, whether a piece of work rests on solid ground or shaky ground — and when work quietly stalls, nothing notices. This has already happened here: a major piece of work was "finished" on paper three months ago, and nobody saw that it was never actually connected to anything.

The proposal: when work begins, turn its opening statement into a small summary the system can read — what the work wants, what is in the way, what it will do, and why. From that, the system can show, from day one, how strong or weak the work looks, what is missing, and what would count as finishing. As the work proceeds, the same readout shows whether it is getting healthier.

What we checked before believing this: we tried it on this very piece of work. It partly worked — the "what to do" and "why" parts came through clearly; the "what we want" part needs better reference material, which we now know how to gather. We also put up the strongest objections we could find, and answered each one:

1. *"These summaries will be built and then ignored, like last time."* Last time, nothing showed that the finished work was never put to use. This time there is a deadline: if the pieces are not in use by then, the work is marked as failed, in a place everyone can see.
2. *"People will write their statements to score well."* The readout is never used as a target or a gate. It is information only, checked occasionally and independently.
3. *"This will make creative work bureaucratic."* The summary describes only the starting point, never the path. The path stays free — that freedom is the point of working this way, and it is kept on purpose.

Every claim above comes with a stated test that would prove it wrong, and the first of those tests has already been run and passed.

Why this is needed, in one line: **work should be able to say what it is for, in a form that lets anyone — or anything — check how it is doing.**

4.2 The argument from the pattern language — a small cascade

The most important version was missing: the ARGUE rebuilt as a **Pattern Language Cascade** — the argument as a composition of *real library patterns*, checked into the arrow store (`futon3a/meme.db`, scope-tag **`cascade/E-mission-head-argue`**, 3 `:constructed` + 3 `:open`).

The demonstrated chain (real patterns, composition shown by this ARGUE):

> `futon-theory/mission-interface-signature` *(work declares a readable, checkable interface — the thesis joint; ≈236 attestations, the corpus's strongest habit)* → `peripherals/read-existing-seam-before-implementing` *(build it by reviving M-aif-head's channels + the exotype bridge — the graveyard answer)* → `realtime/liveness-heartbeats` *(revived parts must emit liveness or die visibly — amendment A2* is *this pattern applied to mission wiring)* → `mission-coherence/logic-model-before-code` *(falsifiers stated before infrastructure: F1–F4 authored first; F1 already ran — the thin-signal answer)*

The three coverage gaps (`:open` candidate-pattern arrows — the argument needs them; the library lacks them):

1. **`candidate-pattern/two-projections-of-one-quantity`** — prose HEAD and compiled sigil as two projections of one intent. *Audit finding:* this was cited as `structure/two-projections-of-one-quantity` in M-memes' PSR — a PSR leaning on a pattern that **was never minted**. The cascade exposes it.
2. **`candidate-pattern/measure-never-target`** — amendment A1 (readouts are information, never gates or optimization targets); recurs across peradams, health readouts, attestation — undocumented as a pattern.
3. **`candidate-pattern/stable-theme-enables-free-improvisation`** — the core defense against Gantt-ification; load-bearing in two missions, not in the library.

Per the grounded-learning loop's own economics (C-falsifiable-missions §7 A3), coverage-gap patterns are the **highest-information units in the store** — so the cascade's gaps are not weaknesses of the argument but its most valuable output: three candidate patterns, each born with a use-site and a rationale. This is the cascade→sorry→wiring discipline applied to an *argument* rather than code — more work checked into the store, as intended.

5. VERIFY (opened 2026-06-10) — two derivations of one argument

ARGUE exit (Joe, 2026-06-10): ratified — the four-voice ARGUE (bout + amendments + plain-language + pattern cascade) passes out of ARGUE into VERIFY.

V1 — the convergence probe (the centerpiece; per combining-methods-as-diagnostic)

Our §4.2 cascade was **designed** top-down from the bout. claude-3's **Build 1** (`futon3a/holes/labs/M-memes-arrows/cascade_construct.py`, phylogeny-grounded, reviewed PASS — C-falsifiable-missions) **assembles** cascades bottom-up from the prior distribution over the pattern library. Run Build 1 over this mission's hole ($\psi \approx$ "make the mission HEAD a typed, checkable object") and **diff the two derivations**. The disagreement is the diagnostic:

- **P1 (recoverability):** the assembled cascade ranks the four designed patterns (`mission-interface-signature`, `read-existing-seam...`, `liveness-heartbeats`, `logic-model-before-code`) high. If yes → the designed argument is *recoverable from the prior* — not idiosyncratic; the pattern language supports it. If no → either the assembler is blind here or the argument leans on unattested structure. Both are findings.
- **P2 (the gap test — sharpest):** the assembler **cannot propose** the three `:open` candidate patterns (they are not in the library, hence not in the prior). Prediction: the diff shows holes exactly at the three gap joints. *Unless* it surfaces near-neighbors that could serve — in which case the candidates shrink or die before minting. This is how candidate patterns earn their flexiargs: survive the near-neighbor search.
- **P3 (the duals):** Closure 01 showed assembly over-selects (26 patterns, 6 phylogeny-orphans); Closure 06 notes design under-covers (4 nodes). The diff measures both vices on one hole — calibration data for Build 2's proposal-scoring (C-falsifiable-missions §7 A2: the cascade is a proposal distribution; score its ranking of the used set).

Form: follow-up excursion (candidate name `E-cascade-convergence`), cross-referenced to claude-3's Build 1 — it verifies this mission AND calibrates their assembler; one run, two consumers. Build 1 may still be in motion; the excursion waits on its owner's nod, not on this mission.

V2 — the standing obligations (already armed)

- **A2's wire-or-die clock**: the three `:open` arrows in `diagram/E-mission-head` (health-readout, seeded-beliefs, lifeform-lane) — promote within two working sessions or the lifeform self-declares `:moribund`. This is VERIFY's liveness check, running by construction.
- **F2–F4** (in `E-mission-head.aif.edn`): decorative-readout, uninformative-sigil, intent-not-typeable — spec-only falsifiers awaiting their runs; F4's run is the HEAD-grain-anchors experiment (§3.4 car 3).

V0 — the logic model: VERIFIED (2026-06-10)

Joe's steer: don't wait on Build 1 — the established VERIFY is the core.logic model, and the pattern specifying it (`mission-coherence/logic-model-before-code`) is **already the fourth pattern in this mission's own ARGUE cascade**. The argument prescribed its own verification; running it:
`futon3c/src/futon3c/logic/mission_head_invariants.clj` — five structural invariants over an abstract lifeform trace:

invariant	encodes	adversarial caught
`measure-never-target`	A1: no gate/optimize decision consumes the health readout	✓
`wire-or-die`	A2: at the horizon, no `:open` arrow while claiming `:alive`	✓
`mode-crossing`	promotion requires a construction; attestation never crosses	✓
`sigil-provenance`	sigils derive from HEAD sections; hand-set bits forbidden	✓
`falsifier-first`	every construction's claim has an *earlier* falsifier	✓

`run-verify` ⇒ `:verified? true` — conforming witness 0 violations; all 5 adversarial fixtures caught by their own category (including the subtle one: a falsifier authored *after* its construction is caught by ground-step comparison, not just absence).
V1 (the Build-1 convergence probe) stays open as the *empirical* complement — optional, parallel, on claude-3's timetable; V0 verifies the design now.

PUR — Pattern Use Record

- Pattern: `mission-coherence/logic-model-before-code`
- Actions taken: encoded the mission's 5 design invariants (A1, A2, mode-crossing, sigil-provenance, falsifier-first) as core.logic+pldb over an abstract lifeform-event trace; authored conforming witness + one adversarial fixture per invariant; ran offline.
- Outcome: success — witness clean, 5/5 adversarials caught.
- Prediction error: minimal — one encoding note: the falsifier-*ordering* check resisted pure relational form and needed ground projection (min over falsifier steps); the house exemplar had no temporal-order invariant, so this is a small extension of the idiom worth carrying back to the pattern.

6. INSTANTIATE (opened 2026-06-10) — three arrows, three bells

The INSTANTIATE work *is* A2's three `:open` arrows, dispatched as parallel bell handoffs from **fable-1** (newly Agency-registered) to the three idle codex agents — contract-first so the consumers build against the schema while the producer implements it. The shared contract: **`<mission>.health.json`** (sibling of the mission doc; sigil + health + degraded mode for prose-only HEADs; schema fixed in the bells).

handoff
H1 — health emitter (`head_exotype_probe.py -emit-health`) **H2** — the revive: wire MissionAifHead into the birth (`:on-cycle-complete`, 4-arity `validate-phase-advance`, `seed-i` **H3** — vitals block in the overview panel (♡ conf · xeno% · section marks · reading)

Each bell carries: goal, `:in`/`:out`, acceptance bar, gates (clj-kondo + check-parens + named tests where applicable), the **hard constraints** (no touching the running JVM/Emacs; health never flows into gate/optimize — the V0 `measure-never-target` invariant restated as a handoff term), and **"bell fable-1 back with summary + commit shas."**
Review protocol (fable-1, on each bell-back): read the diff, re-run the gates, check H2 against `mission_head_invariants.clj`'s invariants, state what was checked; fix small findings directly (carve-out b). **On acceptance: `promote!` the arrow in `diagram/E-mission-head` with the commit sha as the construction ref** — the A2 clock is satisfied by exactly these three promotions, and each promotion is a closure event for the fold ledger.

7. DOCUMENT (opened 2026-06-10) — "Anatomy of a Futonic Mission (from the HEAD down)"

Joe's frame: what this session produced is special — the same mission reached from **two perspectives** (the scope reading in `*mission-overview*`; the AIF reading in the lifeform package), a result Daumal would find satisfying in its own right. DOCUMENT = a **preprint**, companion to "A First Proof Sprint": running-example format, this mission doc as primary source, no verbatim repetition. **The missing piece that makes it worth writing: a synthesis of the two readings.**
Skeleton drafted: `futon6/holes/anatomy-of-a-futonic-mission.md`. The §5 synthesis thesis: *the scope reading is the lifeform's observation space; the AIF reading is its generative model; a mission is well-formed exactly when the two cohere — and the coherence is computable* (ghost lines are literally prediction errors against the canon prior; the H1–H3 wiring is the interoception↔exteroception binding — the body schema). Bonus discharge: the synthesis IS the construction for the cascade's unminted `two-projections-of-one-quantity` — publishing mints the pattern. Peradams form at reading-agreements.
DOCUMENT remains open until the preprint is real and Joe calls the close (mission-close is the operator's call); INSTANTIATE bell-backs land first.

6.1 INSTANTIATE closed (2026-06-10) — three bells, three folds, A2 satisfied

All three handoffs landed and passed fable-1 review (the bells rang back but the return path was mis-aimed — fable-1's resume ran in the wrong project store; root-caused via the job receipts, re-grafted with explicit cwd; the *work* was flawless):

- **H1 PASS** (`futon5@17e9b53`): `-emit-health` producer; both artifacts verified incl. degraded mode. The vitals it computed match §3.3.1 exactly.
- **H2 PASS + one review fix** (`futon3c@9c4033b`): the M-aif-head **revive** — `seed-from-health`, `:on-cycle-complete` wired, 4-arity refusal surface, and **A1 implemented as a runtime check-law** (beyond the bar). Fix: `degraded?` now checks the xenotype, not the sigil container (+ contract- shaped fixtures); 5 tests / 21 assertions green.
- **H3 PASS** (`futon3c@5b87192`): `♡ vitals conf 0.33 · xeno 89% · ⊗IF ⊗HOWEVER ✓THEN ✓BECAUSE` — rendered live in `*mission-overview*` from the H1 artifact. The lifeform-viewer has its lifeform.

All three arrows `promote!`d `:open → :constructed` with commit shas as construction refs — `diagram/E-mission-head` now reads **7 constructed / 0 open. A2's wire-or-die clause is satisfied with the clock barely started; the lifeform keeps `:alive` honestly.** Three fold records added (the H2 one credits `read-existing-seam-before-implementing` — the revive was the fold). Residue for claude-4's seam: `promote!`'s CH2 substrate-2 emission wants a `:payload` on the arrow — promotions succeeded, secondary emission skipped.

V1 — RUN (2026-06-10, via claude-3's construct_cascade review): VALIDATED, with one discovery

Ran the assembler over two ψ phrasings of this mission's hole (fable-1 review of Build 1, claude-3 handoff). **P1:** 3/4 designed patterns recovered — `mission-interface-signature` ranked **#1 in both runs**; the argument is recoverable from the prior. **P2:** all three gap candidates **survive** the near-neighbor search, *enriched*: `war-machine/ideal-actual-gap` = a relative (not substitute) for `two-projections-of-one-quantity`; `ants/baseline-cyber-ant` = the exemplar for `stable-theme-enables-free-improvisation`; nothing near `measure-never-target`. **The divergence finding:** `realtime/liveness-heartbeats` is unrecoverable from either ψ — even phrased as "liveness or visible death" — so the designed cascade's seam→liveness edge is a **missing cross-cluster phylogeny edge** (realtime↔mission-coherence): the first concrete structure-discovery target of the A3 kind. V1 paid rent in both directions: it verified this mission AND handed the assembler a real calibration case (the F1 fork call: B′, per-pattern connectivity gating).